CR23532 – CYBER AND DIGITAL FORENSICS

CASE STUDY ASSIGNMENTS – II

B.E. CSE – Cyber Security | Rajalakshmi Engineering College

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ASSIGNMENT – 2(1)

Case Study: Suspected Unauthorized Access to a College Server

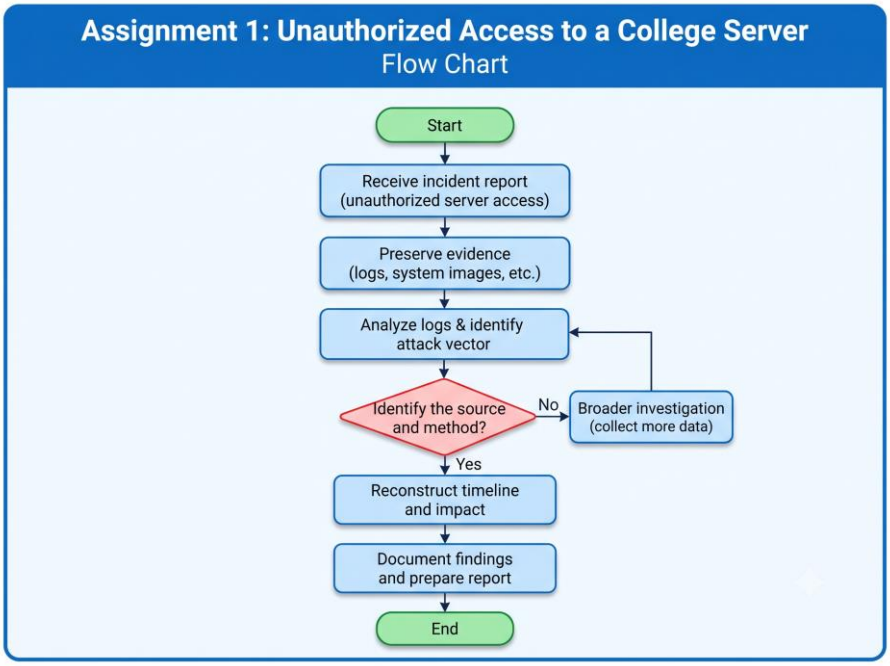
1. Introduction and Objective

A college administrator reports that confidential examination-related files were accessed and copied from a departmental server outside normal working hours. The available sources are the complaint, a suspect workstation, server access logs, a USB device recovered from the workstation, and relevant files. The objective is to determine what activity occurred, when it occurred, which accounts and devices may be involved, and whether the evidence supports the allegation. The investigation must preserve original evidence and avoid assuming guilt before analysis.

2. Forensic Phases

Phase	Activities in this case	Expected output
Identification	Record the complaint, scope, affected server and files, relevant time period, systems, accounts, and persons with knowledge. Confirm written authority and identify volatile evidence that may be lost.	Case scope, evidence-source list, authorization and collection plan.
Collection	Photograph and document devices; isolate systems where appropriate; capture volatile data if justified; acquire server logs and storage using approved methods; use write blockers for removable media where applicable; record handlers, times, and transfers.	Evidence items, acquisition notes, hashes, chain-of-custody records.
Examination	Work from verified forensic copies. Extract logs, file metadata, USB artifacts, account activity, shortcuts, recent-file records, and relevant file contents. Recover deleted material only where appropriate.	Structured artifact set and examination notes.
Analysis	Normalize timestamps; correlate server access, workstation activity, USB connection events, account sessions, file metadata, and hashes. Test alternative explanations and distinguish facts from inferences.	Evidence timeline, correlation matrix, supported findings and limitations.
Presentation	Prepare a clear report with methods, tools and versions, hashes, results, limitations, and exhibits. Explain conclusions in plain language and retain reproducible notes.	Forensic report and exhibits suitable for authorized decision-makers.

3. Investigation Flowchart



4. Diagram

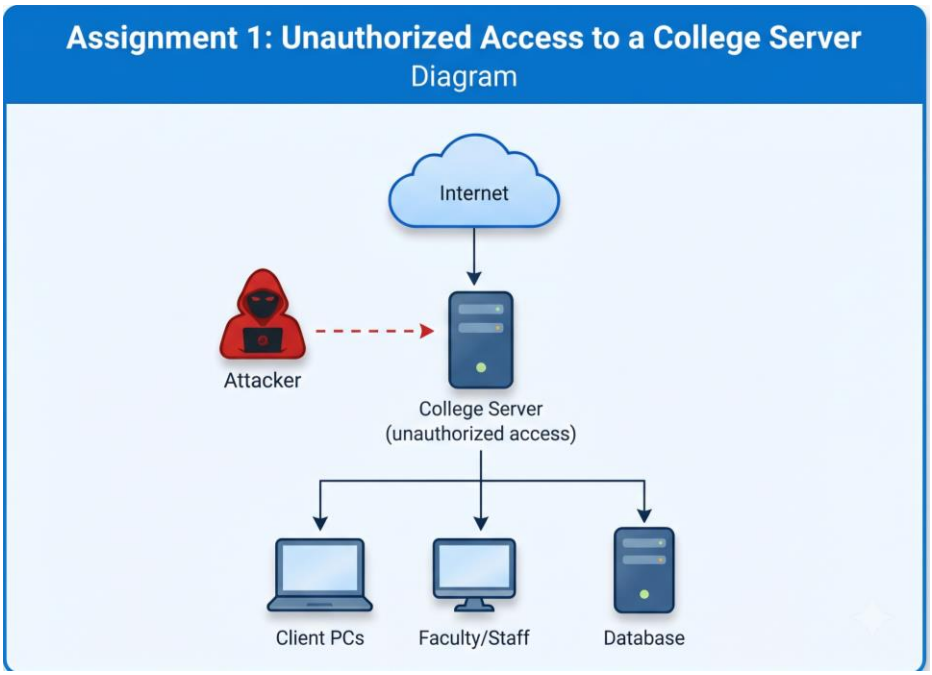


Figure 1 – Digital Forensics Investigation Architecture

5. Potential Evidence and Investigative Value

Evidence item	Potential value	Limitations / handling
Administrator's complaint	Defines allegation, affected files, approximate time window, and initial leads.	A report is an investigative lead, not proof. Preserve original message/document and record who supplied it.
Server access and authentication logs	May show account names, source IPs, access times, file requests, and successful or failed logins.	Check log completeness, clock settings, retention, and whether records can be altered.
Suspect workstation storage	May contain user profiles, recent-file artifacts, browser records, scripts, cached files, and login traces.	Acquire using approved forensic procedures; do not browse the original casually.
Recovered USB device	May contain copied files, file-system metadata, volume identifiers, and connection traces on the workstation.	Image the device with write protection where possible; preserve its physical condition.
Relevant examination files	Can be compared with files on the server or USB using hashes and metadata.	A matching hash supports identical content, but does not by itself prove who copied it.
Account and access-control records	May help determine which accounts had permissions and whether credentials were used.	Account use does not automatically identify the human operator.
CCTV / access-control records, if authorized	May corroborate physical access to the room or workstation.	Synchronize times and assess coverage gaps; collect only within authorized scope.

6. Collection and Preservation Precautions

- Obtain written authorization and define the systems, accounts, date range, and data categories in scope.
- Assign a unique evidence ID to each item. Photograph and document the device, ports, cables, condition, date/time, location, and collector.
- Maintain a chain-of-custody record for every transfer, examination, and storage event.
- Preserve volatile information when justified because memory, active sessions, and network connections can disappear after shutdown. Record the reason and method for any live response.
- Use a validated acquisition method and write blocker where appropriate. Do not connect suspect media to an ordinary workstation for casual inspection.
- Calculate cryptographic hashes, such as SHA-256, for acquired images and relevant exported evidence; verify the image after acquisition and record tool versions and settings.
- Keep original evidence secured and perform examination on verified working copies. Restrict access and document any unavoidable changes.
- Preserve original logs and export context, including source system, export method, time zone, retention policy, and administrator who supplied them.

7. Building an Event Timeline

Investigators should extract timestamps from server logs, authentication logs, workstation artifacts, USB connection records, file-system metadata, and any authorized physical-access records. Record the original timestamp, source, time zone, clock offset if known, and normalized time in a common reference such as UTC. Correlate events by account, device identifier, file path, file hash, source address, and sequence. A hypothetical sequence might be: account login → examination file accessed on server → USB connected to workstation → file-copy artifact created → USB disconnected. Each step must be supported by actual evidence; the sequence above is illustrative, not a finding in this case.

Time (normalized)	Source	Observed event	Confidence / caveat
T1	Authentication log	A user account authenticated to the server.	Confirm account, session, and clock accuracy.
T2	Server access log	A request for a protected file was recorded.	Confirm path, action result, and log completeness.
T3	Workstation USB artifacts	A USB device with a recorded identifier was connected.	Connection does not prove that a particular file was copied.
T4	USB image / file metadata	A file with a matching hash or related metadata was found.	Compare content and metadata; consider other ways the file arrived.
T5	Combined sources	Events align within a plausible time window.	State uncertainty and competing explanations.

8. Legal and Ethical Considerations

The team should follow applicable Indian law and institutional policy, obtain proper authorization, preserve privacy, and collect only data relevant to the authorized investigation. Access to personal accounts or unrelated student/staff information should not exceed the approved scope. Evidence should be handled impartially, securely, and confidentially. Investigators should maintain accurate notes, disclose limitations, avoid unsupported accusations, and preserve exculpatory as well as inculpatory material. If the matter may lead to disciplinary or legal proceedings, the institution should consult its authorized legal or compliance personnel regarding applicable requirements.

9. Sample Forensic Report Structure

1. Case title, case ID, requesting authority, examiner, and dates.
2. Executive summary written for non-technical readers.
3. Allegation, scope, questions to be answered, authorization, and assumptions.
4. Evidence inventory, identifiers, condition, chain of custody, and hash values.
5. Methods, tools and versions, acquisition details, time-zone handling, and validation steps.
6. Findings organized by evidence source, with supporting exhibits and timeline.
7. Correlation and interpretation, including alternative explanations.
8. Limitations, missing evidence, and questions that remain unresolved.
9. Conclusion stated in proportion to the evidence; distinguish observed facts from inference.
10. Appendices: hash manifest, log excerpts, screenshots, timeline, and custody records.

10. Conclusion

A defensible conclusion requires more than finding a suspicious file or a late-night login. The strongest assessment comes from independently corroborated evidence, reliable acquisition, a transparent timeline, and consideration of alternative explanations. The report should explain what the evidence supports and what it cannot establish.

ASSIGNMENT – 2(2)

Case Study: Linking a Suspicious File to a User Using File-System Evidence

1. Objective

A shared desktop contains a suspicious executable, hidden files, encrypted documents, browser artifacts, and multiple user accounts. The aim is to examine the hardware and software environment and assess whether the available artifacts reliably link file activity to a particular account or person. Attribution should be based on multiple independent artifacts rather than a single filename or timestamp.

2. System Inventory

Component	Details to document	Why it matters
Computer hardware	Make/model, serial number, CPU, RAM, internal disks, connected peripherals.	Identifies the system and guides acquisition.
Storage devices	Device model/serial, capacity, interface, partitions, health indicators.	Helps identify evidence locations and acquisition method.
Operating system	Name, edition, build, installation details, configured time zone.	Provides context for artifact locations and timestamp interpretation.
File systems	NTFS/FAT/exFAT or other types, volumes, allocation status.	Determines available metadata and recovery methods.
Accounts and permissions	Local/domain accounts, SIDs where available, profiles, groups, login history.	Helps distinguish account context and access rights.
Applications	Browser versions, security tools, archive/encryption utilities, relevant forensic tools.	Explains artifacts and possible application-generated records.
Network / external devices	Network configuration, mapped drives, USB history, attached media.	May identify additional sources and alternate file-transfer paths.

3. How the Environment Helps Locate Evidence

- Storage devices may retain allocated files, deleted entries, unallocated space, partitions, and application data.
- Operating-system artifacts such as user profiles, recent-file lists, link files, event logs, prefetch data where available, and registry records may show that a file or application was accessed.
- File-system metadata may include file paths, sizes, timestamps, ownership or security identifiers, and allocation status. Metadata must be interpreted with the file system's behavior in mind.
- Hidden-file attributes can reduce visibility in normal file browsing but do not necessarily make content inaccessible to forensic tools.

- Encrypted documents may reveal filenames, sizes, timestamps, containers, or associated application traces even when content cannot be read without a key.
- Browser artifacts such as history, downloads, cached records, and session data may provide context, but shared devices, synchronization, private browsing, and cleanup tools can complicate interpretation.

4. Diagram

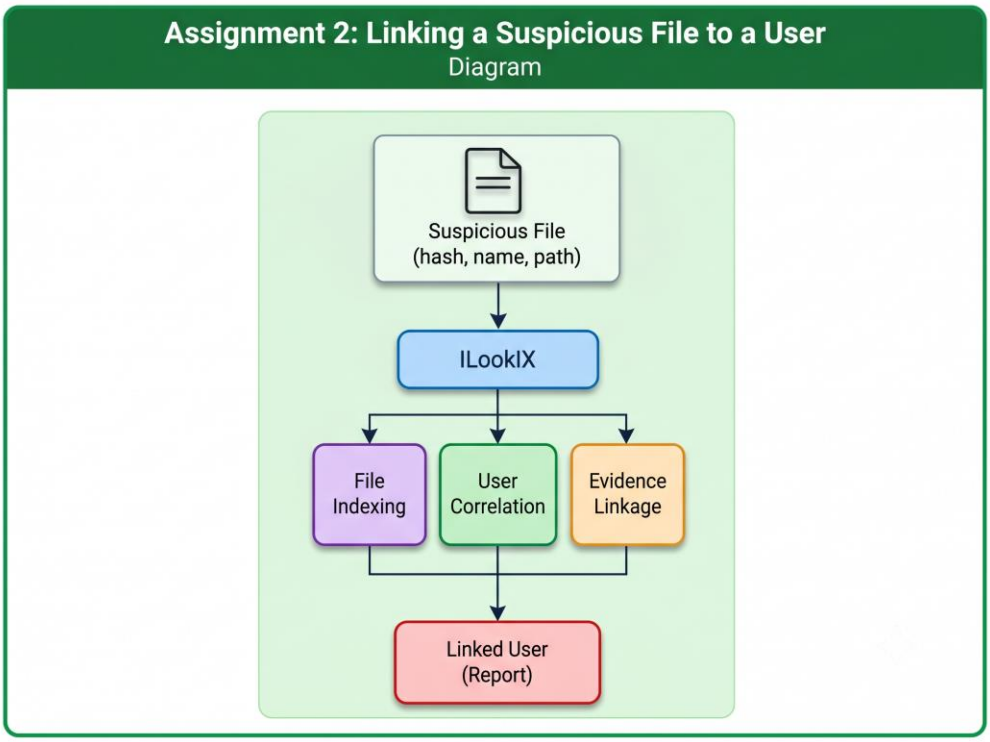


Figure 2 – Forensic Environment and Evidence Linkage

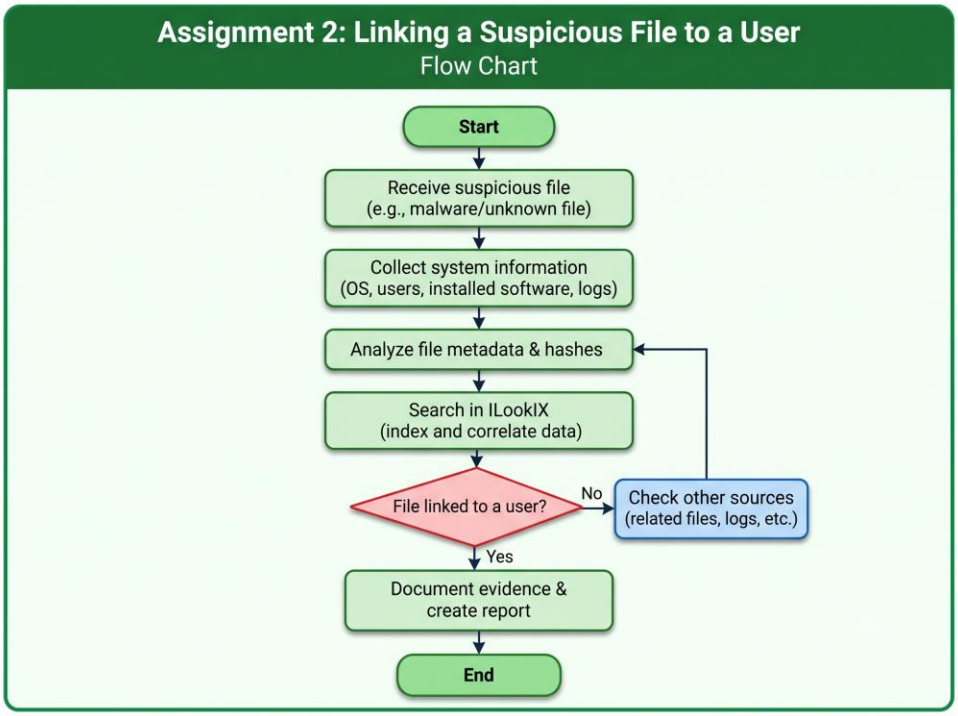
5. Hidden and Encrypted Files

Investigators should record the file path, attributes, size, timestamps, hashes, and the method by which each file was found. They should not modify attributes or attempt decryption on the original evidence. Work from a verified image, preserve the original encrypted content, and use only legally authorized, documented techniques. If decryption is unsuccessful, report the limitation rather than implying that the file's contents are known. Hidden or encrypted status alone is not proof of wrongdoing.

6. Evidence-Linkage Matrix

Artifact	Possible relationship to user	Strength	Cautions
File in a specific user profile	Shows location within a profile associated with an account.	Contextual; stronger with other account-specific artifacts.	A profile may be shared, copied, or accessed by an administrator.
File-system timestamps and path	May help establish when a file was created, modified, or accessed and where it was stored.	Supporting evidence.	Copying, extraction, clock changes, and file-system behavior can change timestamps.
Shortcut / recent-file artifact	May indicate that a file was opened from a particular path in a user context.	Supporting; potentially strong when corroborated.	Artifact may be stale or generated by background activity.
Application or OS event log	May show logon, process execution, or application events associated with an account.	Potentially strong when reliable and corroborated.	Logs may be incomplete, overwritten, or reflect service/background activity.
Browser download record	May link a download URL, time, and local destination.	Supporting evidence.	Browser profiles can be shared or synchronized; downloads can be automated.
USB connection artifact	May show that a particular device was connected to the computer.	Device-use evidence, not proof of file transfer by itself.	Device could have been used by another person or for unrelated tasks.
Matching file hash on another device	Shows that the file contents are identical if a reliable cryptographic hash matches.	Strong evidence of content identity.	Does not independently identify the person who copied or created it.
Login/session records	May place an account in an active session during the relevant period.	Supporting evidence.	A logged-in session does not prove who physically operated the computer.

7. ILookIX Forensic Workflow



8. Direct vs Supporting Artifacts

A direct artifact is one that records a specific action or event in a relatively explicit way—for example, a reliable process-execution record associated with a user session. A supporting artifact provides context, such as a file located in a user profile or a USB device connection. Even a direct record may identify an account or process rather than the human who controlled it. Confidence increases when independent sources agree.

9. Limitations and Alternative Explanations

- The computer is shared, so another staff member may have used the logged-in account.
- Credentials may have been shared, stolen, or used remotely.
- System clocks may be inaccurate or timestamps may be recorded in different time zones.

- Files may have been copied from another computer, restored from backup, synchronized, or generated automatically.
- Logs or artifacts may be missing, overwritten, altered, or affected by privacy/cleanup software.
- Encryption may prevent content examination, and the file name alone may be misleading.
- An antivirus scanner, indexing service, backup agent, or other background process may have touched a file.
- Attribution may be weakened if the evidence image or acquisition integrity cannot be verified.

10. Conclusion

A defensible user-linkage conclusion should identify the exact account, device, file, and time-related artifacts relied upon and state the degree of confidence. No single artifact should be treated as conclusive proof of a person's actions. ILookIX can support structured examination, but the examiner must validate results, preserve provenance, and clearly report limitations.

ASSIGNMENT – 2(3)

Case Study: Forensic Imaging and Safekeeping of a Suspect Storage Device

1. Objective

A laptop and attached USB drive are seized in a suspected data-theft investigation. The purpose is to acquire a defensible forensic copy, preserve the original media, verify integrity, conduct any necessary recovery, and maintain complete custody records. The workflow below describes an intended procedure using IXImager and an ASB container; exact functions and supported options must be confirmed in the installed product documentation.

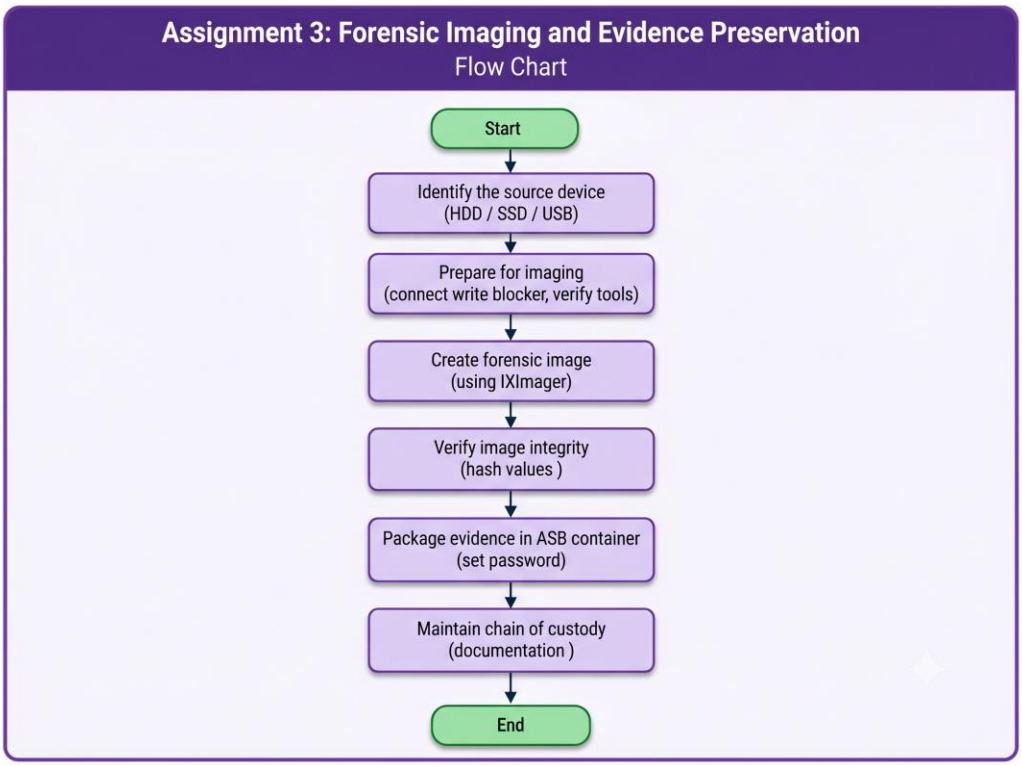
2. Collection and Transport

11. Confirm legal/institutional authority, identify the collection scope, and assign unique evidence IDs.
12. Photograph the laptop, USB drive, screen state, cables, ports, labels, and surrounding setup. Record date, time, location, collector, and device condition.
13. Do not explore files or casually power the device on. If the system is running, consult the approved live-response procedure because shutdown can destroy volatile evidence or trigger encryption.
14. Record whether the laptop is powered on/off, logged in, connected to networks, or showing signs of encryption. Document the decision and rationale for isolation or shutdown.
15. Label the USB drive and document its connection to the laptop before disconnecting it, if safe and appropriate. Package each item separately in suitable anti-static and tamper-evident packaging.
16. Protect devices from shock, moisture, magnetic or electrical hazards, and unauthorized access. Seal, label, and transport them securely.
17. Start a chain-of-custody record and obtain signatures for every handover.

3. Physical Acquisition and Use of a Forensic Image

Physical acquisition aims to capture the accessible addressable contents of a storage device, including allocated data and potentially unallocated space, subject to device, encryption, and acquisition limitations. A forensic image allows repeatable examination without repeatedly interacting with the original. It also supports verification, independent review, and preservation. Where full physical acquisition is unavailable—for example, due to hardware encryption, damaged media, or a mobile-style storage interface—the examiner must document the actual acquisition type and its limitations.

4. IXImager Imaging Workflow



5. Diagram

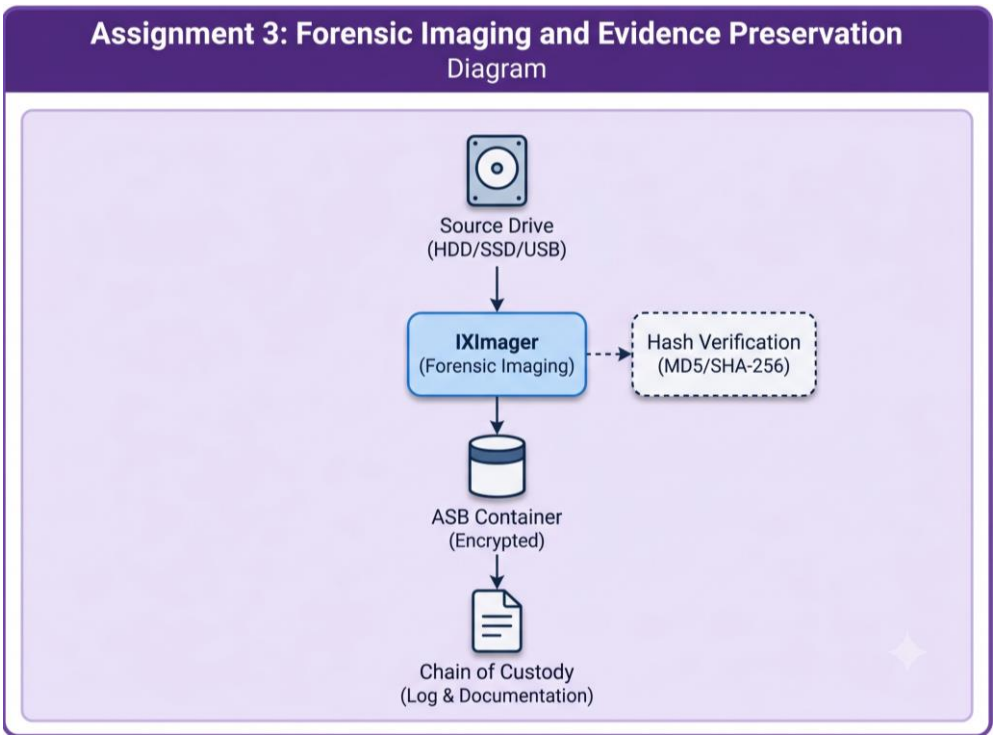


Figure 3 – Forensic Imaging and Evidence Preservation Process

6. Hash Values and Integrity

A cryptographic hash produces a fixed-length digest of data. If the same algorithm is applied to identical data, the resulting digest should match. A matching hash supports the conclusion that the compared data is unchanged with extremely high confidence; it does not prove that the original data was truthful or that a particular person created it. The report should record algorithm, value, tool/version, date/time, object hashed, and verification result. Hashes should be recalculated when copies are transferred or when integrity needs to be checked.

Stage	Record required	Purpose
Before acquisition, where feasible	Source identifiers and any source hash supported by the method.	Establish baseline or explain why a baseline is unavailable.
During acquisition	Tool log, settings, read errors, start/end time.	Make the process reproducible and transparent.
After acquisition	Image hash and verification result.	Check the acquired image against the acquisition record.
During transfer	Hash of transferred copy and receiving-person confirmation.	Check integrity during movement between locations.
Before analysis	Working-copy hash compared with master image hash.	Confirm the analysis copy matches the verified master.

7. ASB Container and Evidence Documentation

An ASB container is referenced in the case as a format for storing or managing forensic evidence. The exact meaning, features, and supported metadata depend on the product or specification used in the laboratory, so they must be confirmed from the relevant documentation rather than assumed. A container may organize acquired data and associated case information; it should not be treated as a substitute for verification or chain of custody.

- Case ID and evidence ID; source device make/model/serial and capacity.
- Acquisition date/time, examiner, location, method, tool name/version, and settings.
- Container/image format and segmentation or compression settings, if used.
- Hash algorithm and values, verification status, and any errors or unreadable areas.
- Source condition, write-protection method, and any live-response or shutdown decisions.
- Custody history, storage location, access permissions, and transfer records.
- Known limitations, encryption status, and references to acquisition logs or appendices.

8. Data Carving

Data carving is the recovery of files or fragments by examining raw data patterns, signatures, or structural features without relying solely on intact file-system records. It may be useful when file-system metadata is missing or damaged, or when relevant data exists in unallocated space. Carving can recover incomplete fragments, lose original filenames and paths, or produce false positives. Recovered items should be stored separately, hashed where appropriate, validated by checking file structure/content, and reported with the method and limitations.

9. Date and Time Problems

- Document the source device's configured time zone, daylight-saving settings if relevant, and clock offset where it can be established.
- Preserve original timestamps and record any normalized time separately; UTC can be used for cross-source comparison.
- Identify timestamp type, such as creation, modification, access, log event, or application-specific time.
- Consider clock drift, manual clock changes, copying/extraction, backup restoration, synchronization, and differences among file systems and applications.
- Do not infer an exact sequence where timestamp precision or clock reliability does not support it. Explain uncertainty in the report.

10. Sample Chain-of-Custody Table

Date/time	Evidence ID / item	Released by	Received by	Purpose / location	Seal / condition / signatures
2026-10-10 09:00	LAP-01 laptop	Collector A	Evidence custodian B	Sealed and transferred to secure storage	Seal no. 101/Intact/Signed
2026-10-10 09:05	USB-01 USB drive	Collector A	Evidence custodian B	Sealed and transferred to secure storage	Seal no. 102 / intact / signed
2026-10-10 11:30	LAP-01 forensic image	Examiner B	Examiner C	Verified image transferred for examination	Hash verified / signed
2026-10-10 12:00	USB-01 forensic image	Examiner B	Examiner C	Verified image transferred for examination	Hash verified / signed

11. Conclusion

Reliable imaging depends on correct source identification, write protection where appropriate, a documented acquisition, hash verification, secure storage, and an unbroken custody record. IXImager and any ASB container should be used according to their actual documented capabilities. All acquisition errors and limitations must be disclosed.

ASSIGNMENT – 2(4)

Case Study: Recovery of Deleted Evidence from Volume Shadow Copies

1. Objective

A Windows forensic image contains Volume Shadow Copies that may preserve earlier versions of documents and browser-related artifacts deleted or changed after a suspected incident. The goal is to select relevant evidence, classify it, reconstruct events, recover appropriate earlier versions, and validate findings before reporting.

2. Evidence Selection and Classification

Category	Examples	Handling / analytical purpose
User-created files	Documents, spreadsheets, scripts, images, project files.	Compare versions, hashes, metadata, and relevant paths.
System/application files	Windows logs, registry hives, browser databases, application configuration.	Establish system context and application/user activity.
Potentially suspicious files	Unknown executables, scripts, unusual archives, files with relevant names or hashes.	Prioritize based on behavior, provenance, and case relevance; do not infer maliciousness from name alone.
Deleted/recovered artifacts	Deleted directory entries, carved files, earlier versions in shadow copies.	Preserve recovery method and distinguish recovered copies from original live files.
Supporting/contextual evidence	Shortcuts, recent-file records, download history, event logs, backup records.	Corroborate chronology and user/session context.
Irrelevant or out-of-scope material	Unrelated personal or business data.	Minimize collection and exposure; do not include without a legitimate reason.

3. Event Analysis

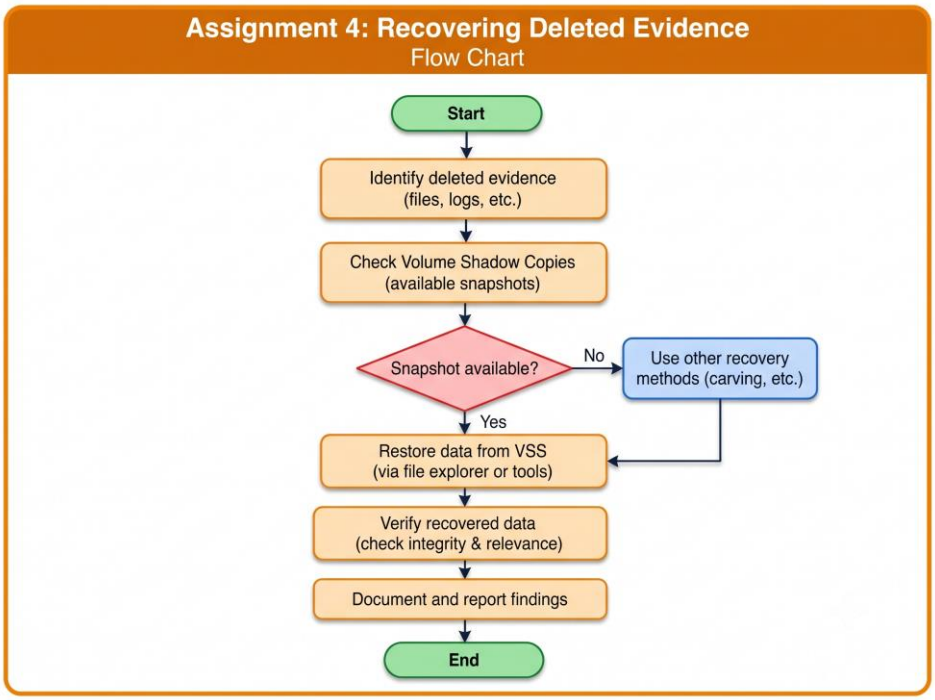
Event Analysis organizes records into a sequence to answer what happened, when, and in what order. Investigators extract events from Windows logs, file-system metadata, browser records, application artifacts, and Volume Shadow Copy versions. Events should be normalized to a common time reference while retaining original timestamps. The timeline should identify gaps, conflicting clocks, duplicate events, and confidence levels. Correlation can support a hypothesis, but temporal proximity alone does not prove causation.

4. Cloud Analysis and Lead Analysis

Cloud Analysis considers authorized cloud records such as sign-in history, file version history, sharing events, synchronization logs, object metadata, and audit records. It may help determine whether a file was uploaded, shared, modified, or accessed from another device. Availability depends on provider retention, account permissions, and the legal scope.

Lead Analysis uses an initial artifact or observation to identify further evidence sources and testable questions. For example, a suspicious file name may lead investigators to search for the same hash, related download records, USB connection events, or earlier file versions. Leads are investigative directions, not conclusions, until corroborated.

5. Volume Shadow Copy Analysis Workflow



6. Diagram

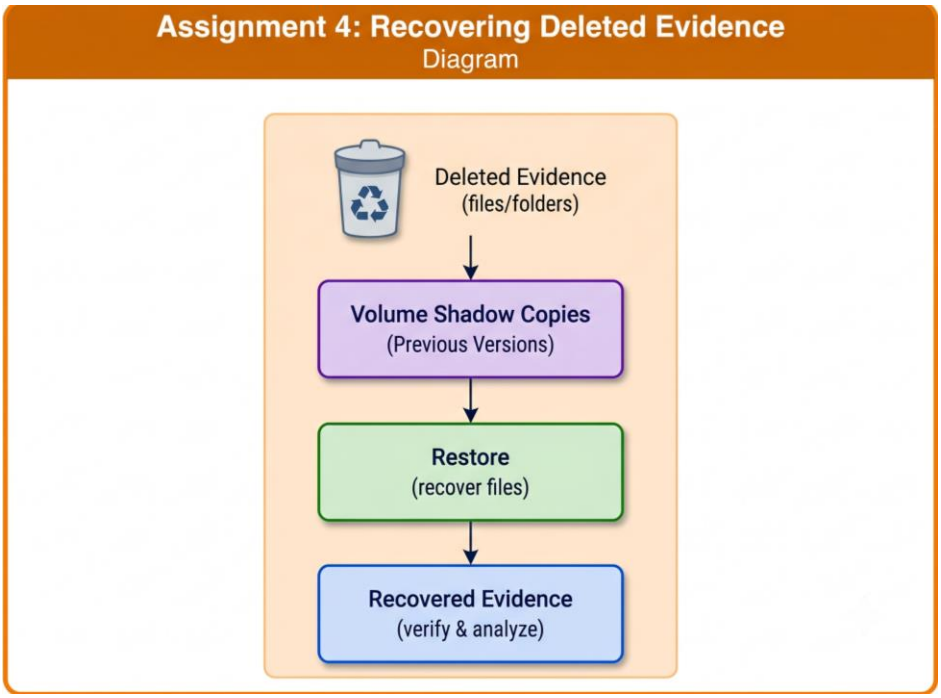


Figure 4 – Deleted Evidence Recovery Using Volume Shadow Copies

7. Hypothetical Timeline

The following example is illustrative only and does not represent actual observations from the supplied case.

Illustrative time	Event	Potential source	Interpretation
09:10	Document created in a work folder.	File-system metadata / earlier snapshot.	Establishes a candidate creation time, subject to clock reliability.
10:25	Document modified.	Metadata / application artifact.	Compare contents and version information.
10:40	Document copied to another folder or removable device.	Recent-file, shortcut, USB or file-system artifacts.	Requires corroboration; no single artifact may prove the transfer.
11:05	Document deleted from the active folder.	File-system records / application logs if available.	Deletion does not necessarily erase all content.
12:00	A Volume Shadow Copy exists containing an earlier version.	Snapshot inventory and recovered file.	May allow comparison with the later state; snapshot creation time is not necessarily the file's original creation time.
Later examination	Earlier version recovered and hashed.	Forensic examination record.	Preserve source snapshot, recovery method, and hash.

8. Validation Checklist

- Confirm the image and working copy hashes match the acquisition record.
- Record the Volume Shadow Copy identifier, volume, and relevant snapshot timestamp.
- Preserve the recovered item's original path and source context.
- Calculate a hash for the recovered artifact and compare versions where useful.
- Check whether content, file structure, metadata, and timestamps are internally consistent.
- Corroborate with at least one independent source where available.
- Document tool/version, extraction steps, errors, and any manual interpretation.
- State alternative explanations and avoid overstating who performed an action.

9. Possible Sources of Error

- File timestamps may reflect copying, restoration, clock drift, or time-zone conversion rather than the event assumed.
- Volume Shadow Copy creation time may differ from the time a particular file was created or changed.
- Some snapshots may be deleted automatically or unavailable because of retention limits.

- Recovered files may be partial, overwritten, or reconstructed from fragments.
- Application caches and browser artifacts may be generated by background tasks or synchronization.
- A deleted file's presence in a snapshot does not prove that the suspect deleted it.
- Tool parsing errors, unsupported formats, and incorrect path associations may lead to misinterpretation.

10. Conclusion

Volume Shadow Copies can provide valuable historical context, but every recovered artifact must retain its source snapshot and provenance. A defensible report compares versions, correlates multiple sources, validates the extraction, and clearly separates observed facts from inferred activity.

ASSIGNMENT – 2(5)

Case Study: Multi-Source Investigation of a Suspected Cyber Incident

1. Objective

A student account is suspected of involvement in unauthorized access to an online service. Authorized evidence sources include a Windows computer, browser artifacts, email and messaging records, a mobile phone, and Internet/cloud records. The aim is to correlate the sources while accounting for differences in time, ownership, synchronization, encryption, retention, and reliability.

2. Evidence from Operating Systems

- Windows: event logs, user profiles, registry artifacts, application execution records where available, file-system metadata, shortcuts, browser databases, USB history, and relevant network configuration.
- Other desktop operating systems: user account records, system/application logs, file metadata, shell or recent-item artifacts, browser data, installed applications, and removable-device records where available.
- Across systems, record OS version, device identity, configured time zone, account identifiers, acquisition method, and known limitations.
- An operating-system artifact often identifies a device, account, or process, not necessarily the person physically operating it.

3. Browser Artifacts and Cautions

Browser evidence may include history, URLs, search queries, downloads, cookies, cache, bookmarks, autofill records, session data, extensions, and browser-profile metadata, depending on the browser and settings. It may help establish access to a service, a download, or the sequence of web activity. However, browser history may be cleared, private browsing may leave fewer artifacts, profiles may synchronize across devices, and background requests may occur without deliberate user action. A URL in history does not by itself prove that a transaction succeeded or that the account owner personally initiated it.

4. Email and Messaging Evidence

- Email: sender and recipient fields, message IDs, dates, headers, routing information, attachments, attachment hashes, folders, and account login/audit records where available.
- Messaging: account identifiers, message timestamps, conversation participants, attachment records, delivery/read states, and device/session information where lawfully available.
- Preserve original exports or native records when possible, along with acquisition method and metadata. Screenshots can be useful exhibits but should not replace underlying records when those records are available.
- Correlate message times and attachments with browser downloads, local files, cloud uploads, and mobile artifacts. Consider forwarding, spoofing, shared accounts, synchronization, and clock differences.

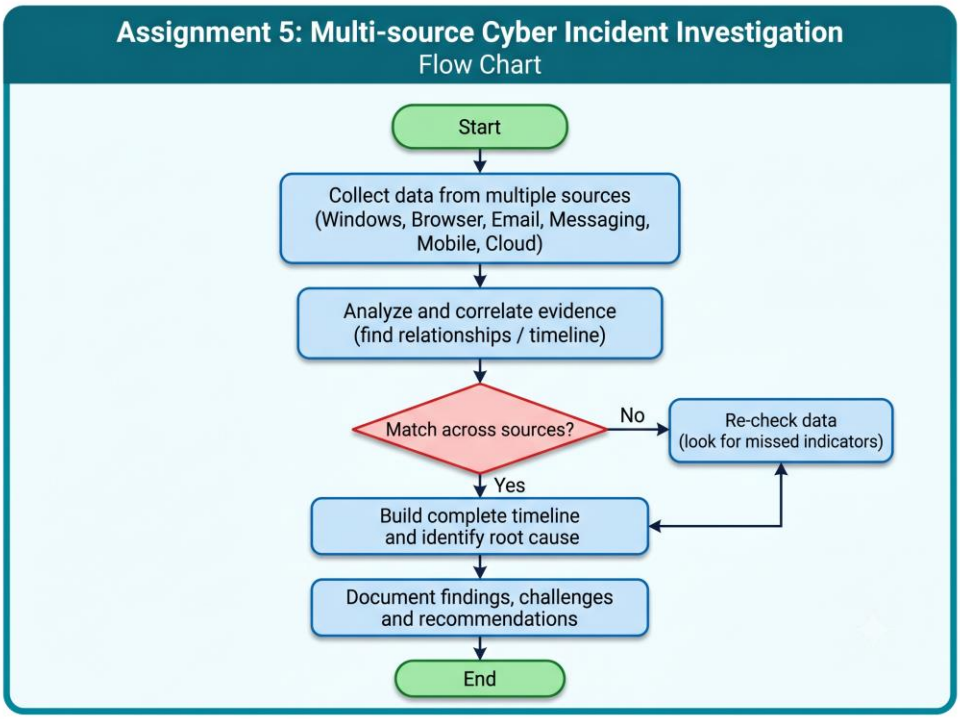
5. Mobile Phone Considerations

Mobile examination should begin with authorization, device identification, condition documentation, and a decision about network isolation and power state. Investigators should consider passcode protection, full-disk or file-based encryption, cloud backups, application-specific storage, remote-wipe risk, SIM/eSIM and account identifiers, and the effect of extraction methods on data. Use a validated and documented acquisition method appropriate to the device and case. Record whether the result is a logical, file-system, backup-based, or physical extraction; these methods do not recover identical data. Avoid claiming access to deleted or encrypted content unless the extraction actually supports it.

6. Internet and Cloud Evidence

- Service sign-in records, IP addresses, session IDs, user-agent information, multi-factor authentication events, and security alerts.
- Cloud file audit logs, file version history, sharing permissions, synchronization activity, object identifiers, and download/upload records.
- DNS, firewall, proxy, VPN, or network records where lawfully retained and relevant.
- Provider records should be collected through authorized channels and accompanied by provenance, retention details, time-zone information, and integrity records where available.
- IP addresses and device identifiers may be shared, dynamic, or masked by VPNs, carrier-grade NAT, proxies, and public networks. They should not be treated as direct proof of an individual's identity.

7. Multi-Source Investigation of a Suspected Cyber Incident Flow chart



8.Diagram

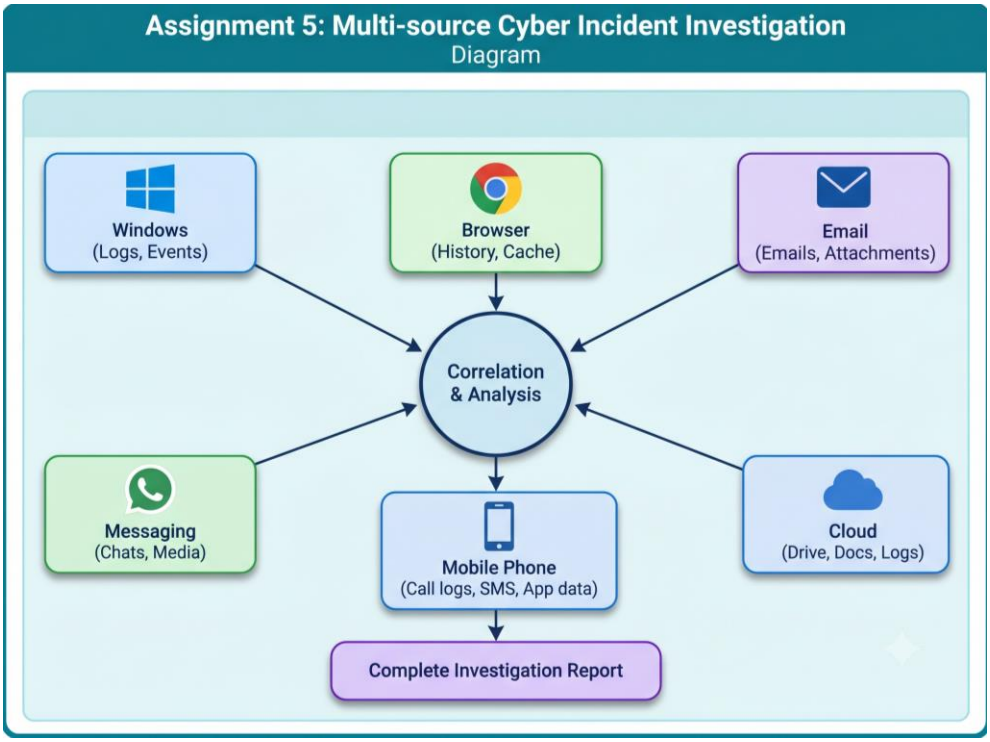


Figure 5 – Multi-Source Digital Evidence Correlation

9. Cross-Source Correlation Matrix

Hypothetical artifact	Source	Possible correlation	Caution
Browser history entry for a service at 20:14	Windows browser profile	Compare with service sign-in logs around the same time.	A history entry may not prove successful authentication.
Service audit record for account login at 20:15	Cloud/service provider	Compare account, IP, session ID, user agent, and authentication result.	Check time zones and whether the record identifies a device reliably.
Email containing a link at 20:05	Email system	Compare message headers and URL with the browser event.	A message receipt does not prove the recipient clicked the link.
Messaging attachment with matching file hash	Messaging record / device	Compare attachment hash with a local file or cloud object.	Matching content does not prove who transferred it.
Mobile app session at 20:16	Mobile extraction / app records	Compare account/session identifiers and device time with provider records.	App timestamps may be local, delayed, or synchronized.
Cloud file download record at 20:22	Cloud audit log	Compare object ID and download time with local file metadata.	A cloud event may be triggered by sync software or another linked device.
USB device connection at 20:30	Windows artifacts	Compare device identifier and relevant file traces.	Connection alone does not establish that the suspected file was copied.

10. Major Challenges in Multi-Source Digital Forensics

18. Different time zones and clock drift: devices and services may record different times. Preserve original timestamps and document normalization.
19. Encryption and locked devices: content may be inaccessible without keys or an appropriate authorized extraction method.
20. Cloud-provider dependence: records may be unavailable, retained briefly, or require formal legal process and provider cooperation.
21. Data volume and heterogeneity: formats, schemas, and application versions differ, making collection and correlation complex.
22. Attribution: shared devices, shared accounts, stolen credentials, VPNs, proxies, and dynamic IP addresses complicate identification of a human actor.
23. Data integrity and provenance: exports, screenshots, conversions, and transfers can lose context or metadata if not documented.
24. Deleted or transient evidence: logs, caches, messages, and volatile data may be overwritten or expire.
25. Privacy and scope: devices may contain unrelated personal information; investigators must minimize unnecessary exposure.
26. Synchronization and duplication: one event may create artifacts on several devices, while a synchronized artifact may not reflect direct user interaction on each device.
27. Tool limitations: parsers may not support every application version or may misinterpret timestamps and proprietary databases.

11. Final Investigation Conclusion

A balanced conclusion should separate direct evidence, corroborating evidence, and unresolved questions. Direct evidence may include a reliable provider audit record showing a successful account event or a well-preserved message record; its significance depends on the record's provenance and what it actually records. Corroborating evidence may include matching browser activity, a local file, mobile session artifacts, and a related cloud event. These artifacts can strengthen a timeline when their identifiers and timestamps align, but none should automatically be equated with proof that a particular student personally performed the activity.

The examiner should state whether the evidence supports, weakens, or cannot resolve the allegation. The report must identify the artifacts relied upon, note time and attribution uncertainties, disclose missing sources, and recommend further authorized collection only where a specific unresolved question justifies it. Because the case description provides no actual logs or extraction results, no factual finding of unauthorized access or personal responsibility can be made from the scenario alone.

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12. Conclusion

A multi-source investigation is strongest when evidence from independent systems converges on the same event and the examiner preserves the provenance of every artifact. Careful normalization, validation, privacy protection, and explicit limitations are essential to a fair and defensible conclusion.

Syllabus Alignment and General References

The assignment structure follows the supplied CR23532 syllabus:

- Unit I – Digital Forensics Process: identification, collection, examination, analysis, presentation, cybercrime law, and investigation methods.
- Unit II – Forensics Environment: hardware/software environments, storage devices, operating systems, file systems, metadata, hidden/encrypted files, and ILookIX.
- Unit III – Collecting Evidences: acquisition, preservation, imaging, date/time issues, data carving, IXImager, and ASB container.
- Unit IV – Analyzing Evidences: selecting and categorizing evidence, Event Analysis, Cloud Analysis, Lead Analysis, Volume Shadow Copy analysis, and validation.
- Unit V – Other Sources of Evidences: operating systems, browsers, email, messaging, mobile phones, Internet/cloud sources, and challenges in digital forensics.